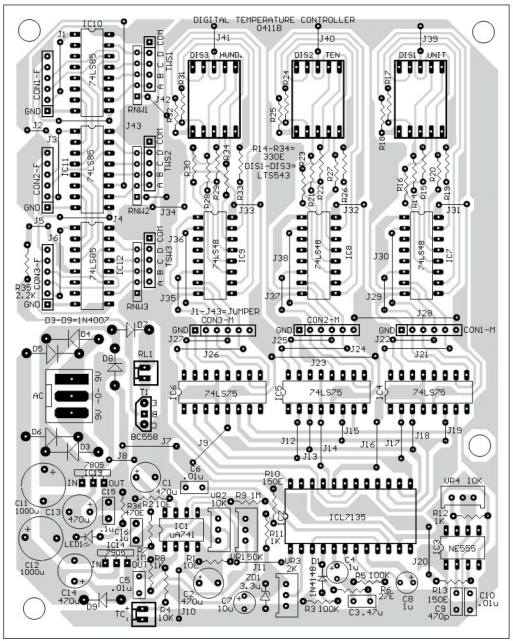


Fig. 6: Component layout for the PCB

remove shorting.

Span setting. Connect the microvolt injector unit through positive and negative leads to pins 2 and 3 of µA741, respectively. Set the oscillator frequency to 120 kHz using preset VR4. Feed the microvolt signal as per the K-type thermocouple chart (given in the EFY-CD).

Preset VR1 connected between pins 2 and 6 of op-amp µA741 is used to get full-scale temperature reading on the display unit according to injection voltage. Preset VR3 is used for reference setting.

After proper calibration, connect the K-type thermocouple between resistor R4 (connected to pins 2 of IC1) and pin 3 of op-amp µA741. It will display the room temperature after connection of thermocouple. Dip the thermocouple in boiling water and check temperature in display. It will display around 100°C.

Relay control setting. Set any temperature through the thumbwheel switch and feed the signal voltage (as per the K-type thermocouple temperature chart) through the voltage injector. When the signal voltage crosses the set temperature voltage, the relay energises to disconnect the heating element of the oven.

EFY note. The controller has been tested up to 300°C. ●

